

ASSER ABDELGAWAD

 Portfolio  GitHub  LinkedIn
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Education

University of Toronto

B.A.Sc. Engineering Science - Electrical and Computer Engineering major, AI minor

Grad Date: APRIL 2027

Toronto, CA

Experience

Hiroshi Ishiguro Labs

MAY 2026 – AUG 2026

Humanoid Robotics Researcher - Voice and Conversation

Osaka, JP

- Built a non-invasive benchmarking framework using realtime traffic capture and voice activity detection.
- Reduced response time by 44% via voice synthesis redesign, LLM inference-stage tuning, and data pipeline optimization.
- Trained voice models to improve the robot's perceived MOS naturalness by 31%.

Tesla Inc.

SEP 2024 – DEC 2024

Low Voltage Test & Automation Intern

Palo Alto, USA

- Created C drivers for port-mapped I/O devices using bit manipulation and shared objects for integration to Python app
- Built brake releaser mechanism using PCAN-Pro dongle and UDS APIs for CAN-FD routines.
- Built CNN using ResNet and OpenCV to detect vehicle features with 99.01% accuracy on production line.
- Designed and conducted full bring-up on an analog front-end and high-side drive PCBA to be used with National Instruments DAQs, and built a matching custom library in Python using the NIDAQ-mx gRPC API.

Tesla Inc.

MAY 2024 – AUG 2024

Electronics Design Engineering Intern

Palo Alto, USA

- Owned MCU signal flow architecture in main controller PCB for the next-gen, high-volume Supercharger program
- Conducted signal integrity tests on 1000BASE-T1 PAM3 Ethernet by quantifying packet error rates on Linux SoC.

Kepler Communications Inc.

SEP 2023 – APR 2024

Power Systems Engineering Intern

Toronto, CA

- Verified Battery Management System's safety logic, control loops, UART buses, and cell voltage monitoring using hardware-in-the-loop test suite with a custom PCBA (design and layout in Altium) and a Raspberry Pi.
- Designed a snubber on a transformer-based isolated flyback converter, DACs, EEPROMs, communication buses, etc.
- Used multithreading for telemetry acquisition (with SMBus library for I²C) and safety controls for a 1.6 kilowatt system.

Pulsenics Inc.

JUN 2023 – SEP 2023

Firmware QA Intern

Toronto, CA

- Developed a PID controller in C for a high-power system involving a power-factor correcting boost converter into a dual active-bridge converter (PFC/DAB) using Code Composer Studio, TI SysConfig, and ARM Cortex-M microcontrollers.
- Verified data acquisition by writing Interrupt Service Routines in C for sensor qualification and ADC configuration.
- Designed Power-over-Ethernet (PoE) switch for 1000BASE-T Ethernet packet communications.

Projects

Custom Software-In-The-Loop Environment

MAR 2025 – MAR 2025

- Built a SIL app for simulating signals to-and-from a temperature sensor over SPI to a bare-metal ARM-based system
- Tested extensively using GTest framework for graceful exits, communication bus failures, and noise filtering solutions

Smart Home Security System

JAN 2025 – FEB 2025

- Used FreeRTOS to process inputs from doppler effect and ultrasonic radar sensors with a queue for inter-task communication, layered atop a custom hardware platform designed in Vivado for a Xilinx SoC.

Biomedical Imaging Software Research

MAY 2022 – AUG 2022

- Programmed signal processing algorithms to deconvolve thermal images, published in 14.7 impact factor journal.

Controllers Lead - UofT Formula Racing

SEP 2021 – SEP 2023

- Designed and manufactured electronic systems in a winning racecar, including system architecture, PCBAs, and firmware

Amazon Customer Review Analyzer - Daisy Hackathon Finalist

JAN 2022 – JAN 2022

- Used Word2Vec machine learning model to analyze Amazon customer reviews for product performance.

Plastania Video Game

NOV 2019 – JAN 2020

- Built an Angry Birds-esque video game in C++ using Allegro graphics package and object-oriented programming.

Technical Skills

Languages & Developer Tools: C, C++, Assembly, Verilog, Python, CMake, MATLAB, Altium, Code Composer Studio

Protocols: CAN, CAN-FD, Quad SPI, I²C, UART, Ethernet, RS-422, RS-485, BLE/UWB, gRPC, TCP/IP, UDP, SCPI

Frameworks & Technologies: FreeRTOS, OpenCV, GTest, PyTest, electronic load, oscilloscope, spectrum analyzer, VNA